

Project Design Phase

Solution Architecture

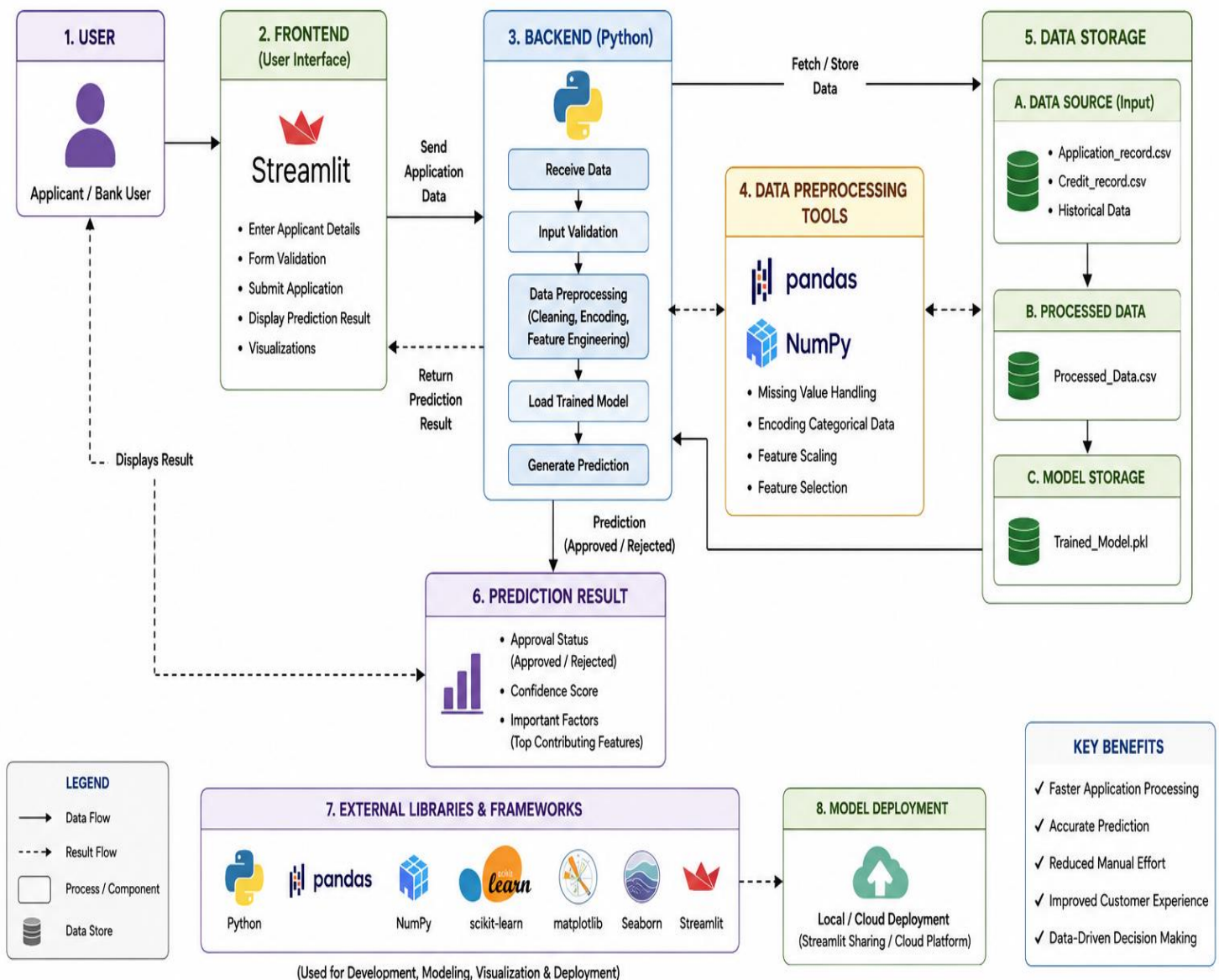
Date	
Team ID	
Project Name	Credit Card Approval Prediction
Maximum marks	

Solution Architecture

The **Credit Card Approval Prediction System** is developed to automate the process of predicting whether a credit card application is likely to be approved or rejected. The system uses machine learning algorithms to analyze applicant information and provide quick, accurate, and reliable predictions.

SOLUTION ARCHITECTURE DIAGRAM CREDIT CARD APPROVAL PREDICTION SYSTEM

(Machine Learning Based Web Application)



Advantages of the Solution

- Faster prediction process.
- Reduces manual verification.
- Improves prediction accuracy.
- Easy-to-use web interface.
- Reliable and scalable system.
- Saves time for banks and customers.

Technologies Used

Component	Technology
Frontend	Streamlit
Backend	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Algorithm	Random Forest Classifier
Visualization	Matplotlib, Seaborn
Model Storage	Joblib
Version Control	Git & GitHub

Conclusion

- The proposed solution architecture combines **Python**, **Machine Learning**, and **Streamlit** to build an efficient credit card approval prediction system. The system automates the approval process, reduces manual effort, and provides quick and accurate predictions, improving the overall decision-making process for financial institutions